

ADVANCE PROBLEM PROGRAMMING

SLO 1: Logic Programming Simulation

1. Simulate an Expert System Rule (age > 18) in Java if-else

An expert system uses rules to make decisions based on given conditions. The rule **“If age is greater than 18, the person is an adult”** can be implemented using if-else.

```
public class Main {  
    public static void main(String[] args) {  
        int age = 20;  
  
        if (age > 18) {  
            System.out.println("Person is an adult");  
        } else {  
            System.out.println("Person is not an adult");  
        }  
    }  
}
```

Output:

Person is an adult

2. Convert Decision Rules into Java Conditions

Decision rules can be represented using Java conditional statements.

Rules:

- If marks $\geq 50 \rightarrow$ Pass
- If marks $\geq 75 \rightarrow$ Good Performance
- If marks $\geq 90 \rightarrow$ Excellent Performance

- Otherwise → Fail

```
public class Main {  
    public static void main(String[] args) {  
        int marks = 85;  
  
        if (marks >= 90) {  
            System.out.println("Excellent Performance");  
        } else if (marks >= 75) {  
            System.out.println("Good Performance");  
        } else if (marks >= 50) {  
            System.out.println("Pass");  
        } else {  
            System.out.println("Fail");  
        }  
    }  
}
```

Output:

Good Performance

Thus, logical rules can be converted into Java conditions using if, else if, and else.

3. Design a Java Quiz App That Uses Rule-Based Logic

```
import java.util.Scanner;  
  
public class Main {  
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
int score = 0;
```

```
System.out.println("Java Quiz");
```

```
System.out.println("1. Which keyword is used to inherit a class?");
```

```
System.out.println("A. implement");
```

```
System.out.println("B. extends");
```

```
System.out.println("C. inherit");
```

```
System.out.println("D. super");
```

```
System.out.print("Enter your answer: ");
```

```
char answer = sc.next().toUpperCase().charAt(0);
```

```
if (answer == 'B') {
```

```
    System.out.println("Correct!");
```

```
    score++;
```

```
} else {
```

```
    System.out.println("Wrong!");
```

```
}
```

```
System.out.println("2. Which method is the starting point of a Java  
program?");
```

```
System.out.println("A. start()");
```

```
System.out.println("B. run()");
```

```
System.out.println("C. main()");
```

```
System.out.println("D. execute()");
```

```
System.out.print("Enter your answer: ");
```

```
answer = sc.next().toUpperCase().charAt(0);
```

```
if (answer == 'C') {
```

```
    System.out.println("Correct!");
```

```
    score++;
```

```
} else {
```

```
    System.out.println("Wrong!");
```

```
}
```

```
System.out.println("Final Score = " + score + "/2");
```

```
if (score == 2) {
```

```
    System.out.println("Excellent!");
```

```
} else if (score == 1) {
```

```
    System.out.println("Good! Keep practicing.");
```

```
} else {
```

```
    System.out.println("Need more practice.");
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

Sample Output:

Java Quiz

1. Which keyword is used to inherit a class?

A. implement

B. extends

C. inherit

D. super

Enter your answer: B

Correct!

2. Which method is the starting point of a Java program?

A. start()

B. run()

C. main()

D. execute()

Enter your answer: C

Correct!

Final Score = 2/2

Excellent!